

Vibro-tactile Feedback for Dial Interaction using an Everyday Object in Augmented Reality

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ABSTRACT

Interaction using everyday objects offer a compelling alternative to gesture and dedicated controller interaction in augmented reality. However, unlike dedicated controllers, everyday objects cannot provide active haptic feedback. To explore active feedback with everyday objects, a within-subjects study was conducted with $n = 28$ participants, using a flowerpot enhanced with vibro-tactile feedback to perform dial interactions and comparing four conditions based on the inclusion and exclusion of audio and haptic feedback. No significant differences between conditions were found for task completion time or accuracy, however measurements of immersion and user preference were higher for the conditions that included haptic feedback.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction devices—Haptic devices, Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality, Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies

1 INTRODUCTION

Designing effective interaction methods for Augmented Reality (AR) experiences remains an ongoing challenge. There are several input mechanisms for interaction available all having different advantages and disadvantages, as an example, while physical hand interaction and gestures can provide natural interaction with virtual elements, these interactions lack any sensation of touch, which can lead to reduced accuracy and lower perception that the virtual content is real. Interaction techniques which utilise everyday objects as physical proxies for virtual objects offer a tactile experience that can provide better task performance compared to free-hand gestures [1]. Everyday objects can be integrated into the user's environment, by classifying objects and assigning them a function [4, 6] or by allowing the user to select objects and assign a function [2], which provides a seamless experience across the physical and virtual spaces. However, interaction using everyday objects lacks one key element: the capacity for active haptic feedback. To explore the impact of active haptic feedback, this paper looks at the addition of active haptic and audio feedback to a particular interaction task utilising an everyday object. Audio feedback was included as it often serves a similar role to active haptic feedback, and can be easily provided through standard AR equipment. As an initial exploration, we focus specifically on the use of an everyday object as a proxy for a virtual dial in a virtual safe opening experience. This research seeks to answer the question: "Does vibro-tactile feedback enhance the use

of an everyday object used as a dial in an AR experience compared to only audio feedback or no feedback?".

2 STUDY DESIGN

To study the addition of vibro-tactile feedback on everyday objects for dial interactions a system was designed using two Eccentric Rotating Mass (ERM) actuators mounted near the opening of a small flowerpot. A Microsoft HoloLens Head Mounted Display (HMD) was used to present the virtual objects to the user, while an OptiTrack motion capture setup was used to track the movement of the flowerpot. The system was built to mimic a rotary combination lock, requiring the input of a four digit code in a continuous sequence, changing turn direction to input each digit.

The ERM motors were programmed to deliver a vibration pulse across the increments on the dial between numbers, mimicking the "click" sensation one might expect from a real safe dial. In conditions that presented audio feedback, this "click" was also presented as an audible effect. Figure 1 shows the study setup; note that in the actual AR display the virtual objects are partially transparent, so users can still see their hand even when occluded by virtual objects.

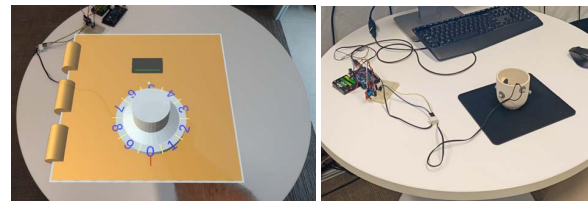


Figure 1: Virtual dial and safe door (Left), and the everyday object and vibro-tactile feedback system (Right) used for the study.

To answer the posed research question a within subjects study was performed with four different feedback conditions: *Audio*, *Haptic*, *Both* audio and haptic, and *Neither* audio nor haptic. The study was designed to test the following hypotheses: H_1 : Users complete a virtual dial code entry task using an everyday object more quickly and accurately if the object is enhanced with vibro-tactile feedback when compared to only audio feedback, or no feedback. H_2 : Vibro-tactile feedback increases immersion when using an everyday object for virtual dial input when compared to only audio feedback, or no feedback. H_3 : Users prefer dial input with an everyday object when it is enhanced with vibro-tactile feedback when compared to only audio feedback, or no feedback.

To test these hypotheses participants used the rotation of the flowerpot to enter a series of four digit codes, three codes were entered for each condition. Participants saw the dial as shown in the left of Figure 1 and received additional feedback depending on the condition. To test for H_1 task completion time and overturn angle (the maximum rotation a participant reached before moving back over the notch for the digit being input) were measured during code entry, for H_2 participants completed the Augmented Reality Immersion (ARI) questionnaire [3] after each condition and finally, for H_3 , participants ranked all conditions from most to least preferred.

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3 RESULTS

A total of 30 participants were recruited for the study, 2 encountered a technical error leading to a lack of vibro-tactile feedback on a condition requiring it. The data from these two participants was excluded from analysis, leading to a dataset of $n = 28$ participants. A demographic survey indicated a balanced gender distribution (14 Female, 13 Male, 1 Other), a young leaning age distribution (15 aged 18-24, 3 aged 25-29, 6 aged 30-34, 1 aged 35-40 and 3 aged 40+) and above average familiarity with using an HMD for AR.

Repeated measure ANOVA was used to test for significance in the differences between conditions for task completion time, accuracy, and immersion factors, using $\alpha = 0.05$ as the threshold for significance. No significant effect was found for either the task completion time, $MSE = 14.63$, $p = .32$, or accuracy, $MSE = 36.1$, $p = .76$.

Figure 2 shows the results for the ARI questionnaire. For this, we consider composite scores for Engagement, Engrossment and Total Immersion based on the responses to questions on each sub-factor [3]. A significant effect of condition was found for Engagement, $MSE = .154$, $p < .001$; Engrossment, $MSE = .295$, $p < .001$, and for Total Immersion, $MSE = .454$, $p < .001$. Tukey HSD tests further showed that for Engagement, the *Neither* condition differed significantly from both the *Haptic* condition and the *Both* condition ($p < .01$ in both cases), but the difference between *Neither* and *Audio* was shy of significance ($p = .059$). For Engrossment and Total Immersion, *Neither* differed significantly from each of the other three condition ($p < .001$ in all cases).

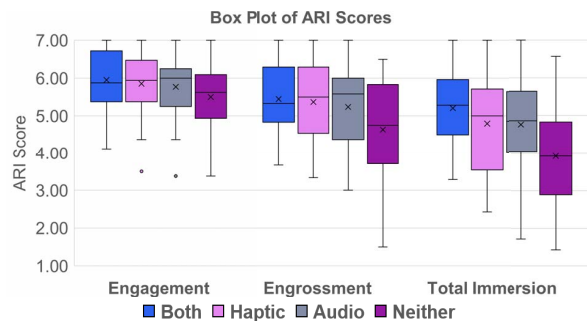


Figure 2: Box plot showing ARI scores for each factor.

Figure 3 shows the distribution of participant preferences for each condition. The *Neither* condition is the least preferred, with 82% of participants ranking this condition last. The *Both* condition was most preferred overall, with 54% of participants ranking it their most preferred and 39% of participants ranking it their second most preferred. Friedman's test was used to compare the mean scores for the ranking data, resulting in a Friedman's Q value = 38.9 with $p < .001$, indicating a significant difference between groups. Further to this Siegel's equation [5] showed that when compared to the *Neither* condition, all other conditions have a significantly different mean rank and that there is a significant difference between the *Both* and *Audio* conditions.

4 DISCUSSION AND CONCLUSION

This study sought to answer the research question: "Does vibro-tactile feedback enhance the use of an everyday object used as a dial in an AR experience compared to only audio feedback or no feedback?". The results of the study provide weak evidence that the answer to this question is yes when compared to no additional feedback but little evidence was found that it provided an improvement over audio. Three hypotheses were tested: for H_1 , concerning task completion time and accuracy, no supporting evidence was found, for both H_2 and H_3 , concerning immersion and condition preference

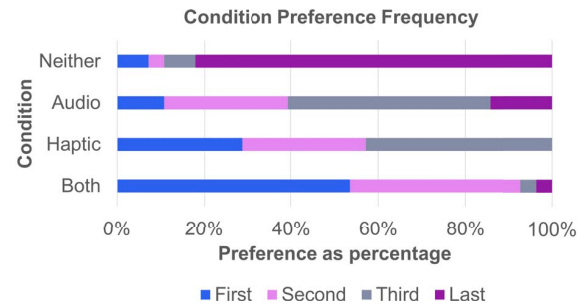


Figure 3: Stacked bar chart showing the frequency that each condition was selected for each level of preference as a percentage.

respectively, evidence was found that conditions with non-visual feedback were better than no feedback (the *Neither* condition), and that participants preferred both forms of feedback to just audio. All together these results provide no evidence that vibro-tactile feedback improves performance elements for dial interaction using an everyday object, however, they do demonstrate that non-visual feedback improves the subjective elements of performing these interactions.

We were surprised to find little evidence that vibro-tactile feedback provided a better subjective experience when compared with audio. There was a significant difference in preference between the *Both* and *Audio*, but not between *Haptic* and *Audio* conditions, and no significant differences existed in factors of immersion between conditions with non-visual feedback. This leads to the conclusion that vibro-tactile feedback provides little to no benefit over only audio feedback when using an everyday object for dial interactions. Therefore designers using everyday objects for similar interactions might want to consider providing only audio feedback to avoid the additional hardware required to provide vibro-tactile feedback.

This study only looked at one form of active haptic feedback and a single type of interaction using an everyday object. In different interaction contexts, vibro-tactile feedback might better enhance the interaction, and different forms of active haptic feedback might better support dial interaction. Further work exploring active haptics and interaction using an everyday object should consider other types of interactions and other forms of active haptic feedback.

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